

English
1001CJA106216230178

Test Pattern


CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
08-04-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

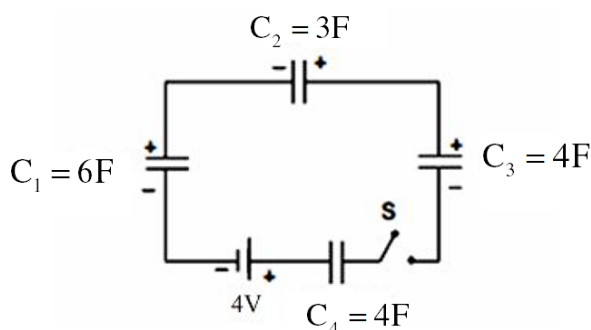
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

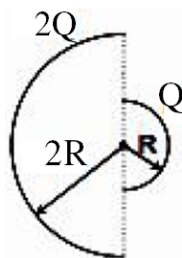
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Three capacitor $C_1 = 6F$, $C_2 = 3F$ and $C_3 = 4F$ are separately charged with battery such that charge on each capacitor are $36C$, $24C$ and $40C$ respectively and then disconnected with the battery. After that they are connected in a circuit along with an uncharged capacitor $C_4 = 4F$ and a battery of emf = 4 volts as shown in figure. Now the switch S is closed. Choose the correct option(s).



- (A) Final charge on capacitor $C_4 = 4C$
- (B) Final charge on capacitor $C_2 = 24C$
- (C) Final charge on capacitor $C_3 = 36C$
- (D) Amount of charge flow in circuit = zero

2. Two hemispherical thin shells made of insulating materials are concentrically arranged in a free space as shown. The radii of the smaller and larger hemispheres are 'R' and '2R' and they carry positive charges 'Q' and '2Q' respectively. The charges are uniformly distributed over the surfaces of the shells. Then choose the correct option(s)



- (A) The electrostatic force between the hemispherical shell is $\frac{Q^2}{8\pi\epsilon_0 R^2}$
- (B) The electrostatic force between the hemispherical shell is $\frac{Q^2}{16\pi\epsilon_0 R^2}$
- (C) The net electric potential at the centre 'O' of the shells is $\frac{Q}{2\pi\epsilon_0 R}$
- (D) The net electric potential at the centre 'O' of the shells is $\frac{3Q}{2\pi\epsilon_0 R}$
3. A uniform solid ball of radius R rolls without slipping between the rails such that the horizontal distance is $\sqrt{3}R$ between the two contact points of the rails with the ball. Figure (a) shows the front view of the ball and figure (b) shows the side view of the ball. V_{CM} is the velocity of centre of mass of the ball and ω is the angular velocity of the ball after rolling down a distance 2h along the incline then

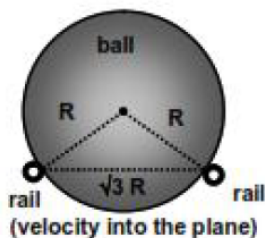


figure (a)

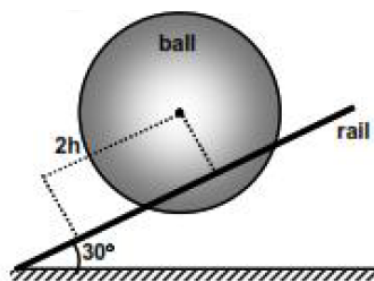
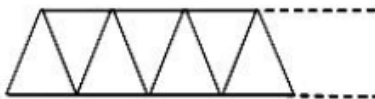


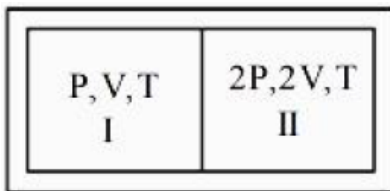
figure (b)

- (A) $V_{CM} = \omega R$
- (B) $V_{CM} = \omega \frac{R}{2}$
- (C) $V_{CM} = \sqrt{\frac{10gh}{13}}$
- (D) $V_{CM} = \sqrt{\frac{10gh}{7}}$

4. n number of identical equilateral prism are kept in contact as shown in figure. If deviation through a single prism is δ . Then (n, m are integers)(Assume surrounding medium be air)

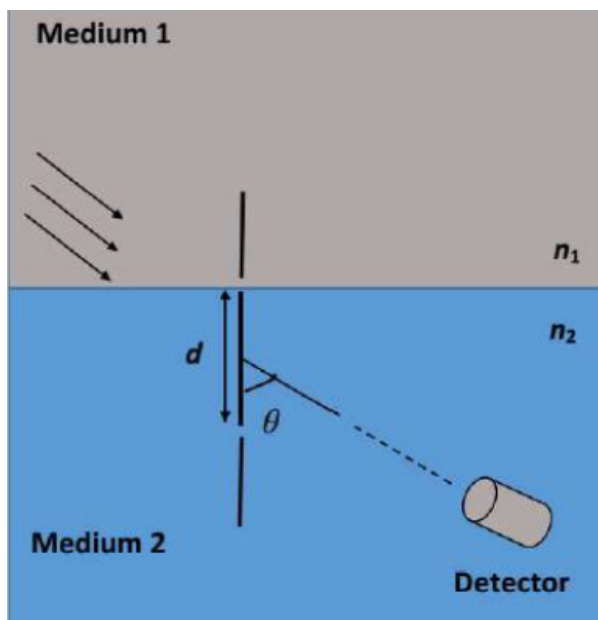


- (A) If $n = 2m$, deviation through n prism is zero.
- (B) If $n = 2m + 1$, deviation through system of n prisms is δ .
- (C) If $n = 2m$, deviation through system of n prisms is δ .
- (D) If $n = 2m+1$, deviation through system of n prism is zero.
5. A partition divides a container having insulated walls into two compartments I and II. The same gas fills the two compartments whose initial parameters are given. The partition is a conducting wall which can move freely without friction. Which of the following statement's is/are correct, with reference to the final equilibrium position?
(initially partition is held at rest)



- (A) The Pressure in the two compartments are equal.
- (B) Volume of compartment I is $\frac{3V}{5}$
- (C) Volume of compartment II is $\frac{12V}{5}$
- (D) Final pressure in compartment I is $\frac{5P}{3}$

6. A double slit setup is shown in the figure. One of the slits is in medium 2 of refractive index n_2 . The other slit is at the interface of this medium with another medium 1 of refractive index n_1 ($n_1 \neq n_2$). The line joining the slits is perpendicular to the interface and the distance between the slits is d . The slit widths are much smaller than d . A monochromatic parallel beam of light is incident on the slits from medium 1. A detector is placed in medium 2 at a large distance from the slits, and at an angle θ from the line joining them, so that θ equals the angle of refraction of the beam. Consider two approximately parallel rays from the slits received by the detector.



Which of the following statement(s) is/are **correct** ?

- (A) The phase difference between the two rays is independent of d .
- (B) The two rays interfere constructively at the detector.
- (C) The phase difference between the two rays depends on n_1 but is independent of n_2 .
- (D) The phase difference between the two rays vanishes only for certain values of d and the angle of incidence of the beam, with θ being the corresponding angle of refraction.

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

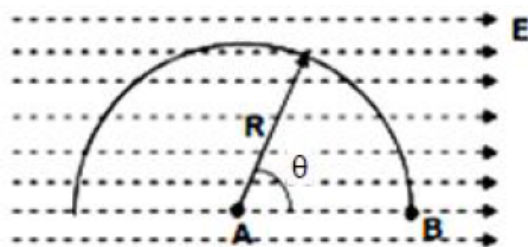
Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

A non conducting semi circular ring of radius R having charge distributed over its circumference according to the formula $\lambda = \lambda_0 \sin\theta$ is placed in a uniform electric field E in such a way that plane of the semi circular ring is parallel to electric field E as shown in figure. Consider that polarizability of the semi circular ring is equal to zero.

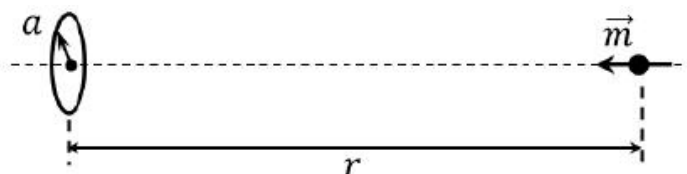


7. Potential at the midpoint of line AB in the absence of electric field E (consider that potential at infinite is zero).
- (A) $\frac{\lambda_0}{\pi\epsilon_0}$ (B) $\frac{\lambda_0}{2\pi\epsilon_0}$
- (C) $\frac{2\lambda_0}{\pi\epsilon_0}$ (D) $\frac{\lambda_0}{4\pi\epsilon_0}$
8. Potential difference between point A and point B in presence of electric field E
- (A) $\frac{\lambda_0}{2\pi\epsilon_0} + E \cdot R$ (B) $\frac{\lambda}{\pi\epsilon_0} + E \cdot R$
- (C) $\frac{\lambda}{2\pi\epsilon_0}$ (D) $E \cdot R$

Paragraph for Questions 9 and 10

A special metal S conducts electricity without any resistance. A closed wire loop, made of S, does not allow any change in flux through itself by inducing a suitable current to generate a compensating flux. The induced current in the loop cannot decay due to its zero resistance. This current gives rise to a magnetic moment which in turn repels the source of magnetic field or flux. Consider such a loop, of radius a , with its center at the origin. A magnetic dipole of moment m is brought along the axis of this loop from infinity to a point at distance r ($\gg a$) from the center of the loop with its north pole always facing the loop, as shown in the figure below.

The magnitude of magnetic field of a dipole m , at a point on its axis at distance r , is $\frac{\mu_0 m}{2\pi r^3}$, where μ_0 is the permeability of free space. The magnitude of the force between two magnetic dipoles with moments, m_1 and m_2 , separated by a distance r on the common axis, with their north poles facing each other, is $\frac{km_1 m_2}{r^4}$, where k is a constant of appropriate dimensions. The direction of this force is along the line joining the two dipoles.



9. When the dipole m is placed at a distance r from the center of the loop (as shown in the figure), the current induced in the loop will be proportional to
- | | |
|---------------------|-----------------------|
| (A) $\frac{m}{r^3}$ | (B) $\frac{m^2}{r^2}$ |
| (C) $\frac{m}{r^2}$ | (D) $\frac{m^2}{r}$ |
10. The work done in bringing the dipole slowly from infinity to a distance r from the center of the loop by the given process is proportional to
- | | |
|-----------------------|-----------------------|
| (A) $\frac{m}{r^5}$ | (B) $\frac{m^2}{r^5}$ |
| (C) $\frac{m^2}{r^6}$ | (D) $\frac{m^2}{r^7}$ |

SECTION-II (i) : (Maximum Marks: 12)

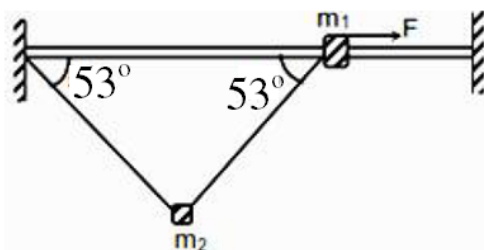
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

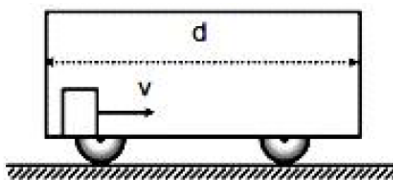
A sleeve of mass $m_1 = 5$ kg can slide on a smooth horizontal rod. A block of mass $m_2 = 2$ kg is suspended at the mid-point of a light inextensible string of length $\ell = 2.3$ m, one end of which is attached to the sleeve and the other end is attached to one end of the rod. Initially the sleeve is held at rest and the angle between the rod and a string segment is $\theta_0 = 53^\circ$ as shown in the figure. A constant horizontal force $F = 20$ N is applied on the sleeve. When the angle between the rod and the string segment becomes $\theta = 37^\circ$, the speeds of the sleeve and the block are v_1 m/s and v_2 m/s respectively. (Take $g = 10$ m/s²)



1. The value of v_1 is _____
2. The value of v_2 is _____

Paragraph for Questions 3 and 4

In a stationary wagon of mass $M = 32 \text{ kg}$ and length $d = 1 \text{ m}$, a small block of mass $m = 1 \text{ kg}$ is projected along its length with a velocity $v = 10 \text{ m/s}$ towards right end from left end as shown in the figure. If coefficient of restitution for all collision is $e = \frac{1}{2}$. All contact surfaces are smooth and each collision is instantaneous.



3. Velocity (in m/s) of the wagon after two collisions is _____
4. Time (in sec) after the projection of the block when the block comes at rest with respect to ground is _____

Paragraph for Questions 5 and 6

A block of ice at 0°C and mass $M \text{ kg}$ is kept inside isolated container of negligible heat capacity. Steam of mass $m \text{ kg}$ at temperature 100°C is kept inside the container. The ice melted and its temperature becomes 20°C in steady state. The steam of mass 0.2 kg at 100°C is again let inside the container, due to which temperature inside the container becomes 30°C in steady state. Latent heat of ice and steam are 80 kcal/kg and 540 kcal/kg respectively. Specific heat capacity of water is $S_w = 1 \text{ kcal/kg}^\circ\text{C}$.

5. The value of m (in kg) is _____
6. The value of M (in Kg) is _____

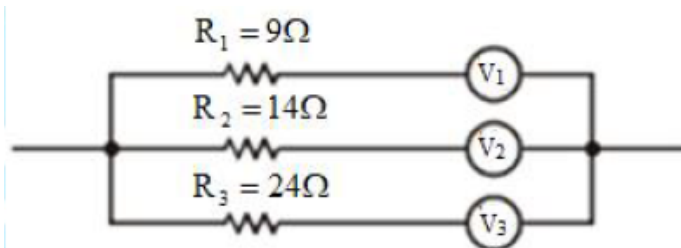
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

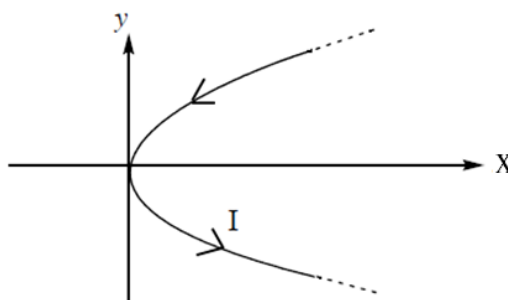
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Consider a hydrogen-like ionized atom with atomic number Z with a single electron. In the emission spectrum of this atom the photon emitted in the $n=2$ to $n=1$ transition has energy 74.8 eV higher than the photon emitted in the $n=3$ to $n=2$ transition. The ionization energy of the hydrogen atom is 13.6 eV. The value of Z is _____
8. The figure shows a portion of an electric circuit. Resistors are known and are indicated on the diagram and the voltmeters are identical. If the voltmeters V_1 and V_2 read 7.5V and 5.0V respectively, find the reading of voltmeter V_3 (in volts).



9. A current $I = 20$ ampere flows through a wire shaped in the form of an infinite parabola of latus rectum $4a$ ($a = 1$ mm). If the magnetic field at the focus of the parabola is $\eta\pi \times 10^{-\beta}$ tesla, where η and β are the positive integers, less than 9. Find the value of $\eta \times \beta$



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

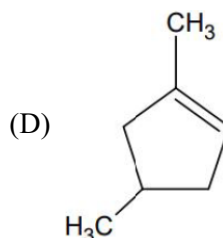
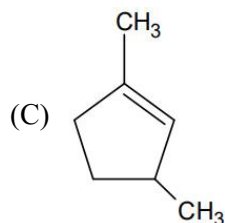
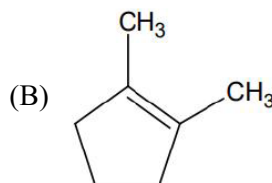
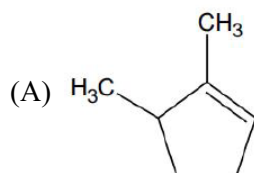
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

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1. Which of the following statements is/are not correct for following compounds
(I) $\text{SCl}_2(\text{OCH}_3)_2$ and (II) $\text{SF}_2(\text{OCH}_3)_2$
- (A) OCH_3 groups in both cases occupy the same position
- (B) Cl — atoms occupy equatorial position in case of (I) and F — atoms occupy equatorial position in case of (II)
- (C) Cl — atoms occupy axial position in case of (I) and F — atoms occupy equatorial position in case of (II)
- (D) Cl and F atoms all occupy axial position in case of both (I) and (II)

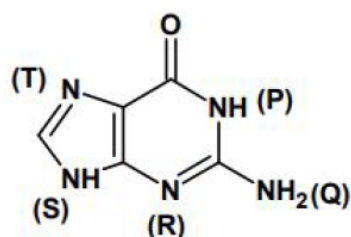
2. Identify the incorrect statement(s) :-

- (A) Combustion of methane in an adiabatic flask will cause no change in temperature of the system
- (B) Dimerization of acetic acid causes an increase in entropy of the system
- (C) It is possible to have both adiabatic reversible and adiabatic irreversible processes between two states
- (D) P - V work is always non-zero when there is a change in volume

3. Which compounds would not give 5-keto-2-methylhexanal upon ozonolysis?



4. Correct statement(s) relative to given base is/are



- (A) Basic strength at (T) is greater than (S)
- (B) pK_b difference between (R) and (T) is greater (in magnitude) than the difference between (P) and (T)
- (C) Hydrogen from nitrogen at (P) is more acidic than hydrogen from nitrogen at (Q)
- (D) Delocalization of lone pair on nitrogen at (S) will bring aromaticity to the smaller ring

5. If the radius of first Bohr's orbit of H-atom is x , which of the following is/are the correct conclusion(s)?
- (A) The de-Broglie wavelength in the third Bohr orbit of H-atom = $6\pi x$
- (B) The fourth Bohr's radius of He^+ ion = $8x$
- (C) The de-Broglie wavelength in third Bohr's orbit of $\text{Li}^{2+} = 2\pi x$
- (D) The second Bohr's radius of $\text{Be}^{3+} = x$
6. For the following options, select all the right combinations of concentration cell and condition for the spontaneity given P_1 and P_2 are gas pressures and C_1 , C_2 are concentration terms with all other term carrying their usual meanings
- (A) $\text{Cl}^- (C_1) / \text{Cl}_2, \text{Pt} || \text{Pt}, \text{Cl}_2 / \text{Cl}^- (C_2); C_1 > C_2$
 $\text{1 atm} \qquad \qquad \qquad \text{1 atm}$
- (B) $\text{Fe}_{(s)} | \text{Fe}^{+3}_{(aq)} (C_1) || \text{Fe}^{+3}_{(aq)} (C_2) | \text{Fe}; C_1 > C_2$
- (C) $\text{Pt} (\text{H}_2) | \text{HCl} | \text{Pt} (\text{H}_2); P_1 > P_2$
 $P_1 \qquad \qquad \qquad P_2$
- (D) $\text{Pd} (\text{H}_2) | \text{HCl} (C_1) || \text{HCl} (C_2) | \text{Pd} (\text{H}_2); C_2 > C_1$
 $\text{1 atm} \qquad \qquad \qquad \text{1 atm}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

A sample (X) of compound(s) contains any one or two of the following AgNO_3 , ZnSO_4 , CuSO_4 , MgCl_2 , CaCO_3 , BaCl_2 , Na_2SO_4 , NaCl , NaOH

When (X) is dissolved in water, a clear solution is obtained. This solution is divided into two parts I and II. Part I turns pink on addition of a few drops of phenolphthalein. When dilute hydrochloric acid is added slowly to part II, a white precipitate (Y) is obtained which dissolves on further addition of dilute hydrochloric acid

7. The clear solution obtained when (X) is dissolved in water, contains

- (A) ZnCl_4^{2-}
- (B) AgCl_2^-
- (C) $\text{Zn}(\text{OH})_4^{2-}$
- (D) CuCl_4^{2-}

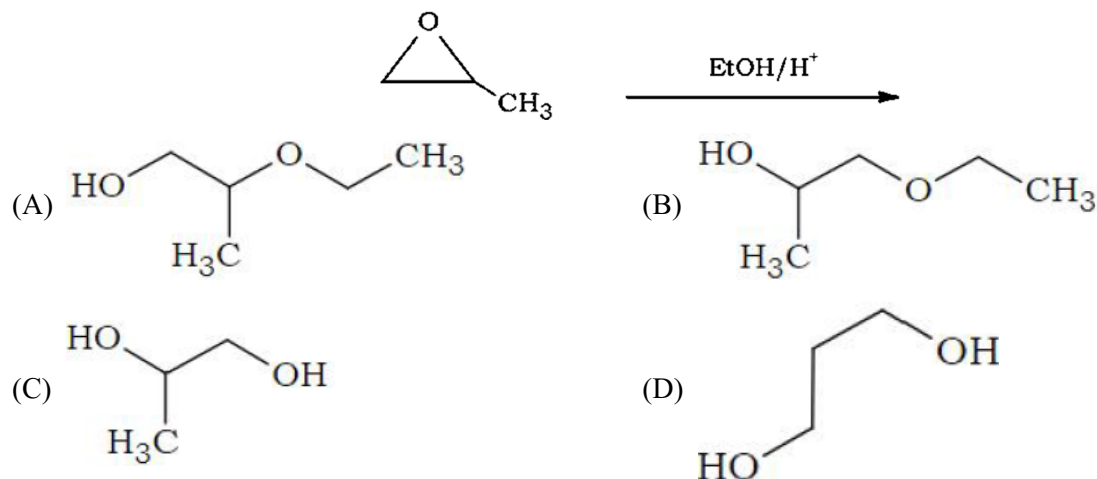
8. The sample (X) contains

- (A) AgNO_3 & NaCl
- (B) ZnSO_4 & BaCl_2
- (C) HgCl_2 & NaOH
- (D) ZnSO_4 & NaOH

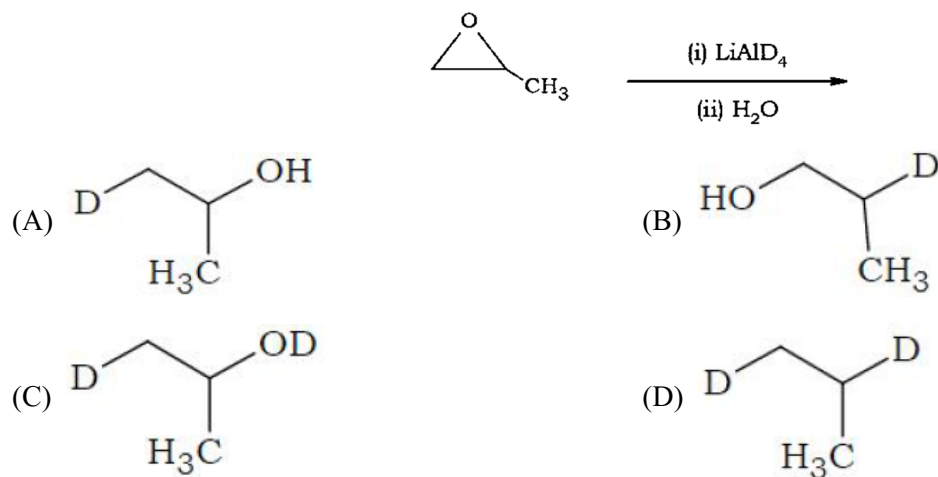
Paragraph for Questions 9 and 10

Due to severe angle strain, epoxides undergo ring opening reactions readily. In the presence of an acid catalyst, such as sulfuric acid, epoxides are hydrolyzed to glycols. Because there is some carbocation character developed in the transition state for an acid-catalyzed epoxide ring opening, attack of the nucleophile on unsymmetrical epoxides occurs preferentially at the carbon better able to bear a partial positive charge. Epoxides undergo ring-opening reactions with a variety of nucleophiles. Good nucleophiles attack an epoxide ring by an S_N2 mechanism and show an S_N2 -like regioselectivity; that is, the nucleophile attacks at the less hindered carbon.

9. The product of the following reaction is



10. The product of the following reaction is



SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

An enclosed glass bulb of volume 'V' contains a pinch of solid NH_4Cl and $\frac{1}{100}$ moles of Ne gas. At 300 K, the internal pressure inside the glass bulb is 114 mm Hg which increases to 908 mm Hg when the bulb is heated to 600 K. (Assume ideal gas behavior, no—sublimation and complete dissociation of NH_4Cl at 600K)

[The only reaction taking place is : $\text{NH}_4\text{Cl}_{(s)} \rightarrow \text{NH}_{3(g)} + \text{HCl}_{(g)}$]

1. What will be the partial pressure of $\text{NH}_3(g)$ (in mm of Hg) in the bulb at 600K ?
2. What mass of NH_4Cl (in grams) is approximately present in 'Pinch' ?

Paragraph for Questions 3 and 4

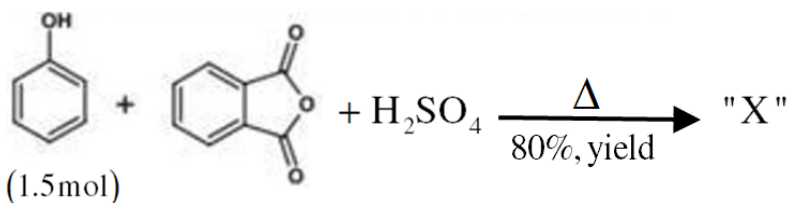
NaClO_3 on reaction with SO_2 gas in presence of H_2SO_4 and trace of NaCl forms a yellow gas (A) as one of the products. Ozonolysis of 'A' produces 'X' along with liberation of oxygen gas.

3. The oxidation state of central atom in 'A' is '+y'. value of 'y' is.
4. 1 mole of 'X' on reaction with excess of NaOH gives 'B', 'Q' and H_2O . Oxidation states of central atoms in the anions of 'B' and 'Q' are '+p' and '+q' respectively. Value of $|q - p|$ is

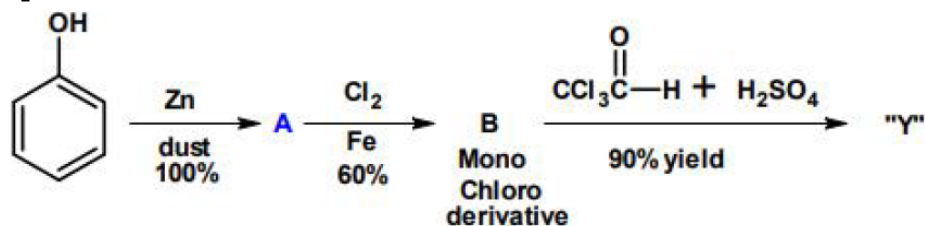
Paragraph for Questions 5 and 6

Phenol is one of the starting materials of two commercially important compounds "X" & "Y". They are prepared as follows.

Experiment I



Experiment II



(1.5 mol)

5. What is the mass (in grams) of 'X' obtained in Experiment - I
6. What is the mass (in grams) of "Y" obtained in Experiment - II

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. In a vessel, two equilibria are simultaneously established at a given temperature as follows:
- $$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) [K_{p_1}]$$
- $$\text{N}_2(\text{g}) + 2\text{H}_2(\text{g}) \rightleftharpoons \text{N}_2\text{H}_4(\text{g}) [K_{p_2}]$$
- Initially the vessel contains N_2 and H_2 in the molar ratio of 9 : 13. The equilibrium pressure is $7P_0$. The pressure due to NH_3 is P_0 and due to H_2 is $2P_0$. If the value of $(K_{p_1} + K_{p_2})$ is $\frac{1}{aP_0^2}$. Find the value of a
8. If the freezing point of 0.01 molal aqueous solution of a cobalt (III) chloride - ammonia complex (which behaves as a strong electrolyte) is -0.0558°C , the number of chloride(s) in the coordination sphere of the complex is $[K_f \text{ of water} = 1.86 \text{ K kg mol}^{-1}]$
9. The number of chloride ions surrounding each Fe^{+3} ion in solid anhydrous ferric chloride is

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $\int e^{x^2} \cdot e^x (2x^2 + x + 1) dx = e^{x^2} \cdot f(x) + C$ where $f(x)$ is some non-zero constant function and C is some arbitrary constant. If the local minimum value of $f(x)$ is equal to m , then:
[[.] denotes greatest integer function]

(A) $f(x)$ is increasing in $(0, \infty)$

(B) the value of $\lim_{x \rightarrow 0} (1 + f(x))^{1/x}$ is equal to 1

(C) the value of $\int_0^1 (f(x) + e^x) dx$ is equal to $2e$

(D) the value of $\left\lceil \frac{-1}{m} \right\rceil$ is equal to 2.

2. Let $f_1(x) = \sin \pi x$, $f_2(x) = \cos \pi x$, $f_3(x) = [x]$, $f_4(x) = \{x\}$, $f_5(x) = -2x$, $f_6(x) = |x|$ be given functions then which of the following statements is/are not correct? (Where $[.]$ is G.I.F and $\{.\}$ is F.P.F., $|x|$)
- (A) Two functions $f_a(x)$ and $f_b(x)$ are selected at random then the probability that $f_a(f_b(x))$ is a constant function ($a \neq b$) is ($a = 1, 2, \dots, 6$; $b = 1, 2, \dots, 6$) $\frac{1}{5}$.
- (B) Two functions are selected at random and if it is known that their sum is a periodic, function, probability that the period is 2 is $\frac{1}{5}$.
- (C) If a function is selected at random from the given functions then the probability that it is a bijection from $\mathbb{R}$ to $\mathbb{R}$ is $\frac{3}{4}$
- (D) If two functions are selected at random and added then the probability that the sum is an even function is $\frac{1}{15}$
3. The function f is continuous and has the property $f(f(x)) = 1-x$ for all $x \in \mathbb{R}$ and $J = \int_0^1 f(x) dx$ then
- (A) $f\left(\frac{1}{4}\right) + f\left(\frac{3}{4}\right) = 1$
- (B) The value of J equals to $\frac{1}{2}$
- (C) $f\left(\frac{1}{3}\right) f\left(\frac{2}{3}\right) = 1$
- (D) $\int_0^{\pi/2} \frac{\sin x}{(\sin x + \cos x)^3} dx$ has the same value as J
4. If $a, b, c \in \mathbb{R}^+$ then $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{(an+k)(bn+k)} =$
- (A) $\frac{1}{a-b} \log \frac{b(a+1)}{a(a+1)}$ if $a \neq b$
- (B) $\frac{1}{a(a+1)}$ if $a = b$
- (C) $\frac{1}{a-b} \log \frac{a(b+1)}{b(a+1)}$ if $a \neq b$
- (D) does not exists if $a = b$

5. Consider the two parabolas $y^2 = x - 1$ and $x = y - 1$. Let A be the area between the coordinate axes, the two parabolas and the line which makes the shortest distance between two parabolas and A^1 be the area between $x = 0$, $x^2 = y - 1$, $x = y$ and the line of shortest distance between $y^2 = x - 1$ and $x^2 = y - 1$ then which of the following is/are true?
- (A) $A = A^1$
- (B) $A = 2A^1$
- (C) The shortest distance between the parabolas is $\frac{3}{2\sqrt{2}}$
- (D) The shortest distance between the parabolas is $\frac{1}{2}$
6. Let $f(x) = x^3 - 3x + 1$ then which of the following can be a solution of the equation $f(x) = 0$
- (A) $2 \cos \frac{2\pi}{9}$
- (B) $2 \cos \frac{4\pi}{9}$
- (C) $2 \cos \frac{8\pi}{9}$
- (D) $2 \sin \frac{\pi}{18}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Let $a_1, a_2, a_3, \dots$ be an arithmetic sequence and $b_1, b_2, b_3, \dots$ be a geometric sequence. Suppose $a_1 b_1 = 20$, $a_2 b_2 = 19$, $a_3 b_3 = 14$. Now the value of $a_4 b_4$ needs to be maximized.

7. The common ratio of the geometric sequence to achieve the maximization is

- (A) $\frac{3}{2}$ (B) $\frac{1}{2}$
 (C) $\frac{7}{5}$ (D) $\frac{1}{3}$

8. The maximum value of $a_4 b_4$ is

- (A) 1.25 (B) 8.50
 (C) 9.25 (D) 9.75

Paragraph for Questions 9 and 10

A ray of light emanating from a point A(1,2,3) and strikes a line minor

$$L : \frac{x+2}{1} = \frac{y-3}{2} = \frac{z-1}{3}$$

at point P, after reflection it passes through the point B(3,6,9).

9. The equation of plane passing through the points P, A and B is

- (A) $x + y - z = 0$
 (B) $2x - y + z = 9$
 (C) $5x - 7y + 3z = 0$
 (D) $4x - 5y + 2z = 0$

10. The value of $PA + PB$ equals

- (A) $\frac{7}{\sqrt{2}}$ (B) $7\sqrt{2}$
 (C) $\frac{49}{\sqrt{2}}$ (D) $49\sqrt{2}$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A_n and B_n be square matrices of order 3 which are defined as $A_n = [a_{ij}]$ and $B_n = [b_{ij}]$ where

$$a_{ij} = \frac{2i+j}{3^{2n}} \quad \text{and} \quad b_{ij} = \frac{3i-j}{2^{2n}} \quad \forall i, j, 1 \leq i, j \leq 3. \quad \text{If } L = \lim_{n \rightarrow \infty} \text{Tr} (3A_1 + 3^2 A_2 + 3^3 A_3 + \dots + 3^n A_n),$$

$$M = \lim_{n \rightarrow \infty} \text{Tr} (2B_1 + 2^2 B_2 + 2^3 B_3 + \dots + 2^n B_n)$$

(Where $\text{Tr} (P)$ = Trace of matrix P). Let $p, n \in \mathbb{N}$,

$$f(x) = 1 - x^p \quad \text{and} \quad g_n(x) = \frac{n}{\frac{1}{f(x)} + \frac{1}{f(2x)} + \dots + \frac{1}{f(nx)}} \quad \text{and} \quad R = \lim_{x \rightarrow 0} \frac{1 - g_n(x)}{x^p}$$

at $n = 5$ and $p = 3$

1. The value of $100 - R$ is equal to _____
2. The value of $L + M + R$ is equal to _____

Paragraph for Questions 3 and 4

Let $f_1(x) = |x-1| + |x-2| + |x-3|$, $f_2(x) = |x-1| + |x-2| + |x-3| + |x-4|$, A = minimum value of $f_1(x)$,

$$B = \text{minimum value of } f_2(x), f(x) = 12x^2 \int_0^{\frac{A}{2}} yf(y)dy + 20 \int_0^{\frac{B}{4}} xy^2f(y)dy + 4x$$

3. The maximum value of $f(x)$, is for $x \in \text{Real}$ _____
4. Area bounded by the curve $y = f(x)$ and x - axis is _____

Paragraph for Questions 5 and 6

If $p \neq 1$ is an n^{th} root of unity having least argument then

$$1 + 5p + 9p^2 + \dots + (4n-3)p^{n-1} = \frac{an}{p-1} \text{ and hence the values of the sums}$$

$$1 + 5 \cos\left(\frac{2\pi}{n}\right) + 9 \cos\left(\frac{4\pi}{n}\right) + \dots + (4n-3) \cos\left(\frac{2(n-1)\pi}{n}\right) = b.n \text{ and}$$

$$5 \sin\left(\frac{2\pi}{n}\right) + 9 \sin\left(\frac{4\pi}{n}\right) + \dots + (4n-3) \sin\left(\frac{2(n-1)\pi}{n}\right) = c \left(n \cot \frac{\pi}{n}\right) \text{ for } n \geq 2$$

5. The value of $a + b + c$ is equal to _____
6. The argument of $z = a - i(b + c)$ is θ (θ is in radians) then $8\theta =$ _____

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Let $g(\theta) = \frac{|\hat{a} \times \hat{b}|^2}{-(\hat{a} \cdot \hat{b})^2} \int (2t+1) dt$ where θ is the angle between $\hat{a}$ and $\hat{b}$. If volume of the parallelepiped whose coterminous edges are represented by the vectors $\hat{a}, \hat{a} \times \hat{b}$ and $\hat{a} \times (\hat{a} \times \hat{b})$ (Where angle between $\hat{a}$ and $\hat{b}$ is taken from the equation $2g(\theta)-1=0$) is $\frac{p}{q}$ then the least value of $p+q$ is _____ mention $\hat{a}$ & $\hat{b}$ are unit vectors where $p, q \in \mathbb{N}$
8. Let $P_k = \begin{cases} k, & \text{if } k \text{ is odd} \\ \frac{1}{k-1}, & \text{if } k \text{ is even} \end{cases}$ and $\prod_{k=1}^{14} (P_k \log_{(k+1)}(k+2)) = N$, then the number of solutions of the equation $\tan^2 x + \cot^2 x + 2 = N \sin^2 2x$ in $x \in [0, 2\pi]$ is _____
9. Let $f(x) = x + x^2 + x^4 + x^8 + x^{16} + x^{32} + \dots$. If the coefficient of x^{10} in $f(f(x))$ is k , then $\frac{k}{5} = \underline{\hspace{2cm}}$

Space for Rough Work

NAME OF THE CANDIDATE

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I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
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paper and ORS codes are the same.

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